PROJECT REPORT

Design and Simulation of a CMOS Ring Oscillator and Analysis of Frequency Variation with Number of Stages

Submitted by

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1 Aim

The aim of this project is to design and simulate a ring oscillator using a combination of CMOS NAND gate and inverters in the DSCH simulation environment, and to study the variation of oscillation frequency with respect to the number of stages.

2 Introduction

A ring oscillator is one of the simplest and most widely used oscillator circuits in digital and analog integrated circuit design. It consists of an odd number of inverting stages connected in a closed loop, such that the output of the last stage is fed back to the input of the first stage. Due to the propagation delay of each inverter, the signal continuously toggles between logic high and logic low, resulting in a periodic square wave output .

In CMOS technology, ring oscillators are commonly implemented using cascaded CMOS inverters. The oscillation frequency of the ring oscillator is determined by the total propagation delay of all stages in the loop. This delay depends on several factors, including the number of stages and transistor characteristics. The relationship between frequency and delay makes ring oscillators very useful for studying the performance of digital circuits and device characteristics.

In this project, the ring oscillator is designed and simulated using the DSCH simulation environment. The transient response of the circuit is analyzed to observe the oscillating waveform and to measure the frequency of oscillation.

In practical designs, a control mechanism is often required to enable or disable oscillation. This is achieved by incorporating a NAND gate at the input of the ring, forming a gated ring oscillator. When the enable signal is active, the NAND gate behaves as an inverter and allows oscillation. When disabled, the output is forced to a constant value, stopping the oscillation.

Ring oscillators are widely used in applications such as clock generation, on-chip testing, process variation analysis, and performance monitoring.

3 Theory

A ring oscillator is a feedback oscillator formed by connecting an odd number of inverting stages in a closed loop. In CMOS implementation, these stages are realized using CMOS inverters. The output of the last stage is fed back to the input of the first stage, forming a loop with no stable logic state. Due to the finite

propagation delay of each inverter, the signal continuously toggles between logic high and logic low, producing a periodic square wave output.

For oscillation to occur, the total phase shift around the loop must be 180° due to inversion, and the loop gain must be sufficient. An odd number of inverters ensures the required phase shift condition, enabling sustained oscillations.

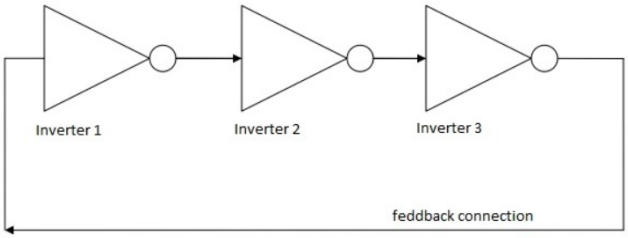

Figure 1: Ring Oscillator Diagram

3.1 Frequency of Ring Oscillator

The total time required for one complete oscillation cycle is equal to twice the delay through all stages. Hence, the oscillation frequency is given by:

$$f = \frac{1}{2nt_p} \tag{1}$$

where n is the number of stages and t_p is the propagation delay of each stage.

3.2 Effect of Number of Stages

From the above equation, it is clear that:

$$f \propto \frac{1}{n} \tag{2}$$

As the number of stages increases, the total delay increases, resulting in a decrease in oscillation frequency.

4 Circuit Diagram

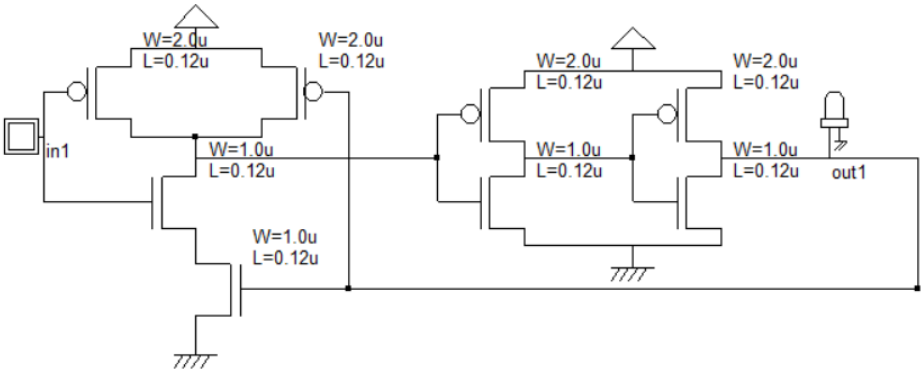

Figure 2: 3-stage Ring Oscillator

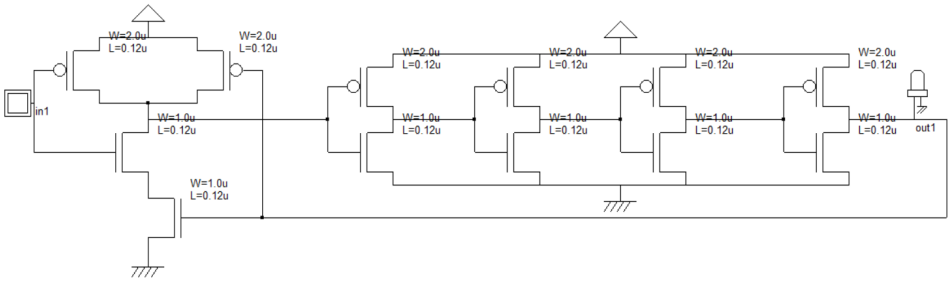

Figure 3: 5-stage Ring Oscillator

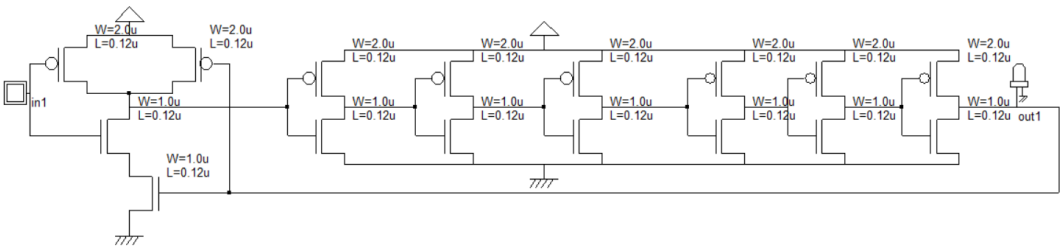

Figure 4: 7-stage Ring Oscillator

5 Simulation Results

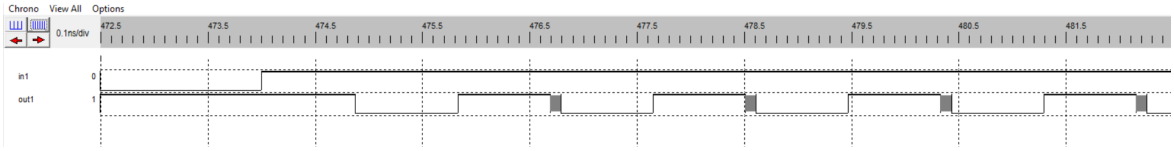

Figure 5: 3-stage Ring Oscillator

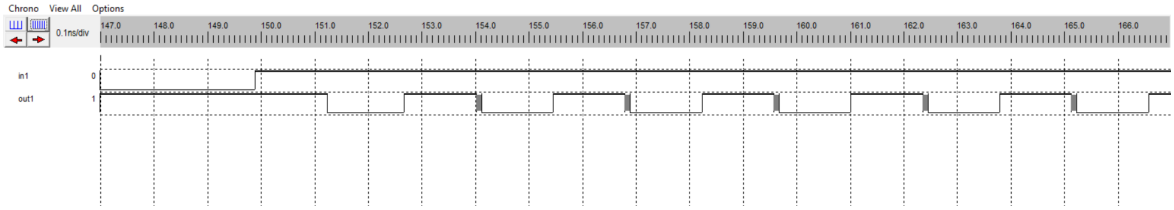

Figure 6: 5-stage Ring Oscillator

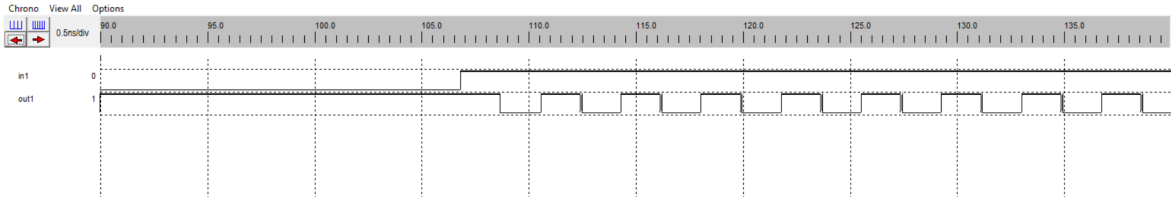

Figure 7: 7-stage Ring Oscillator

6 Result and Discussion

The transient analysis of the 3-stage ring oscillator produces a periodic square wave output, confirming proper oscillation. From the waveform, the time period is measured as:

$$T = 1.7\,\text{ns} \tag{3}$$

The corresponding frequency is calculated as:

$$f = \frac{1}{T} = \frac{1}{1.7 \times 10^{-9}} \approx 588\,\text{MHz} \tag{4}$$

For 5-stage ring oscillator , It is observed from the waveform :

$$T = 2.7\,\text{ns} \tag{5}$$

The corresponding frequency is calculated as:

$$f = \frac{1}{T} = \frac{1}{2.7 \times 10^{-9}} \approx 370\,\text{MHz} \tag{6}$$

Similarly observed Time period and Frequency for 7-stage ring oscillator ,

$$T = 3.7\,\text{ns} \tag{7}$$

The corresponding frequency is calculated as:

$$f = \frac{1}{T} = \frac{1}{3.7 \times 10^{-9}} \approx 270\,\text{MHz} \tag{8}$$

From the simulation results, it is observed that the oscillation frequency depends on the number of stages and the transistor dimensions.As the number of stages n increases, the oscillation frequency decreases. This is because the total propagation delay of the loop increases with the number of stages.

7 Conclusion

The CMOS ring oscillator was successfully designed and simulated using different numbers of inverter stages. The transient analysis produced a stable oscillating square wave, confirming proper operation of the circuit. From the results, it is observed that the oscillation frequency decreases as the number of stages increases. This is due to the increase in total propagation delay in the feedback loop. Hence, the frequency of a ring oscillator is inversely proportional to the number of stages. The simulation results are consistent with theoretical expectations, validating the relationship between propagation delay and oscillation frequency in CMOS ring oscillators.

8 References

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4. Lecture notes and class materials on Ring Oscillators and CMOS Circuits.